

ActInf Textbook Group ~ Cohort 1 ~ Meeting 6

(Chapter 3 pt. 1)

Source: [YouTube](#) Playlist: ActInf Textbook Group ~ Cohort 1 Duration: 1:00:31

| Uploaded: Unknown Transcript method: auto_caption

hello everyone it is june 9 2022 and it's chapter three week one and it's the first cohort of the textbook group we're gonna go to the questions uh there weren't any ideas added for three but if people have any key themes or any ideas that they picked up on with three they can add them here otherwise we'll go to the questions and [Music] also if people have questions in the chat or it would be also great just to take notes or add questions and continue to upload them and then next week we'll continue the discussion on three so this is just the opening set of questions maybe just the first ideas of the book turning two questions now that we've at least gone through chapter two once and part of chapter three or the whole thing of what is the high road to active inference what does anyone think about the high road to active inference or about this low road high road construct that is used in chapters two and three just broadly how did the approach of chapter three feel different than the approach or the topics of chapter two um jf and then ali well i i just want to make the bill comment that the the the the high road approach is what uh first grabbed me when i encountered the free energy principle that that what really fascinated me that there was an attempt at the a unifying answer to uh self-organization uh cognition um and that this could all be brought together uh around some relatively uh a sparse number of of concepts that's what grabbed me much more than the low road thank you ali and then ben well yeah uh reading chapter three actually clarified uh my previously held in fact wrong belief that uh i previously i thought that a low road was a kind of uh bottom-up approach and the high road is a top-down approach but uh now i see that low road was basically uh tackling the problem of active inference from the inferential point of view and from the statistical and inferential point of view but uh in the high road uh we have a much more general generalized and much more let's say all-encompassing uh view to tackle the problem of active inference namely by way of defining markup blankets and all the properties of living organisms and so on awesome thank you ben and then anyone else who raises their hand and also you can take notes on this ideas the book like high road generally

ben and then mike yeah i think uh i think for me as well the height this chapter three in the high road has been um an easier road in and i the thing that i noticed about it was it seems to me so so they say in the introduction that active inference is a normative theory i feel like the high road to me feels there's much more normativity in the high road is that it's an account of what organisms have to do and i think compared to the previous compared to the previous chapter i think chapter 2 felt a lot more descriptive for me i'm not sure other people would maybe that's just the way that i was reading it but it it felt like chapter three was a lot more kind of it had a more normative force to it that i i really enjoyed thank you bye there we go sorry my computer was not responding uh so i went through a similar experience as ali described in terms of going into it thinking about bottoms up and top down and coming out of it thinking about it differently but i feel like in chapter three there's still a lot to unpack around the markov blanket so that is still capturing this inferential nature of things it still has a bayesian quality like what we saw in chapter two but it also for me at least it still feels a bit opaque at this point in terms of this sort of indirect influence that's taking place across the markov blanket awesome okay any other comments otherwise we're going to go to the questions and and we'll approach many of those topics first question regarding the markov blanket explanation on page 43 and box 3.1 they write no additional information about the future is gained by finding out about the past assuming we know the present how can this be true that's one question they write they define regarding the markov blanket as the set of variables that mediate all statistical interactions between a system and its environment what is a statistical rather than a non-statistical interaction they write that they have supplemented conditional independencies with dynamical constraints so that the flows do not depend upon states on the opposite side of the blanket why does independence between the flows on the two sides of the blanket matter so much all right great so who would like to give a thought on this first question how can it be true about markov blankets that no additional information about the future is gained by finding out about the past so i added a note in there you can tell me if this is correct or not with regard to markov chains that as i understand the markov chain in its present state is effectively capturing all the information about the past so i assume that translates into the markov yes um great comment and a markov chain is is a used very broadly just to give one example it's like the current state of the chess board knowing more about previous moves doesn't tell you more about the future and so it's like it's a one markov chain because the dependency is only about that one time step now if it were to violate that condition of being a one markov blanket through time and have like a three markov blanket

that would mean that the same board position depending on what move happened three before was still having some influence on how the system evolved that wouldn't violate the markovian property it would just make it a different kind of markov chain and that is commonly used in the temporal modeling like in time series modeling and in like transitions of stochastic processes the markov blanket is abstracting it from this like time sequence idea like the now is a blanket between the past and the future and we could think about spatial systems like a newton's cradle or something like as well as like spatial temporal systems or just abstract causal systems um so how can this be true it's true by definition because that's what the definition of a markov blanket is whether that description is adequate or maps on to carving nature of the joints and all these other things are um respectively statistical model comparison and ontological philosophical questions but it's true by definition about markov blankets lyle yeah i mean i think you actually you just pretty much covered it but i would just say this is the same condition if you're building rl models of you know different kinds of autonomous systems this is the same restriction i think your comment about the nested the nested blankets is interesting because in the rl space you know we we typically don't have that kind of richness but what we do then would be for example we want to know that just and in that case we would just create a you basically create a state variable which is the the pro you know the the history of the uh of the state transition so we fudge it a little bit but your your explanation that in in in this conception with nested markov blankets then that that would be the more comprehensive solution to that issue thanks just to kind of clarify there like one hack or work around would be like you make the list of all of the last possible three moves and then you make a one markov blanket with all the last combinations of three so it still uses the machinery of a one markov chain which has the simplest transition matrix but it could encompass the last states but it sort of compresses them into a one markov chain format yeah i mean very often you want to know the you know cross you want to know the lat the average of the last you know 20 time steps yeah you're just trying to smooth something right and it's just way too much work to build all that machinery in there so you you make you make that average actually a present variable is what you do great example if something really does depend on the last 20 time steps or you want to do model comparison as to whether it's a better model that is conditioned on the last 15 or 20 or 30 um you can just condense it with summary statistics and descriptions again into this one markov framework otherwise the combinatorics of how the 20 influence each other is vast brock i guess that's related to it like my comment was just going to be that um the way they worded it if you're not rigidly thinking about us

singular markov blanket you know i mean if just practically speaking finding out about the past usually does give you information about the future if you're talking about outside of a singular marketplace right so um if you you know adding the words um no additional information like uh about the mark of you know about the future of the markov blanket is gained by finding out about the past the markov blanket like kind of you know bounds that but um in the case of like the machine learning example and comparing models you're talking necessarily talking about a much larger archive blanket than just a singular one there right so that's just the same thing yes and just like many concepts have like a broader conversational informal sense and a narrow more technical sense assuming we know the present means fully knowing the presence fully knowing the blanket so it's like but i could still learn things about the past so that's not to say that there isn't novel information that you could discover about the past like you could still go back in that chess game and learn information so it's not that there isn't information in the past it's not that it wouldn't even be interesting it's that for the purposes of how the present goes into the next time step the present blanket in this temporal markov chain which is like a blanket through time it contains the information you need for the transition frequencies um okay they define that it's the set of variables that mediate all and also the markov blanket um using last names you know personal opinion is not helpful because not only are there multiple markovs but the markovian property is not something that's technical or defined maybe someday we'll have better ways to describe and be more specific that don't involve invoking ambiguity around names that being said it was worked out analytically by markov and sun and others but most recently it was pearl 1988 and causal inference with bayesian perspective so this is not like an act of inference ism it's drawing upon a total bayesian statistical framework any base graph that's not fully connected is going to have some nodes that intermediate so that's going to relate to this question so what is a statistical rather than a ali and then anyone else who would like to address that well i think it has something to do with the way uh these interactions are parameterized and modeled because you see every phenomenon can be parameterized and modeled uh from many different many various perspectives frameworks within various frameworks but here and the reason i think they put the term statistical in parentheses is that they mean that these variables mediate all interactions that have been parameterized statistically so the i think it's it refers to the nature of parameterizing and modeling these interactions as opposed to distinguishing between inter statistical and non-statistical so anyone else can raise their hand but just here's a few thoughts on that um a statistical interaction is the edge in a causal bayes graph and people might be

more familiar or have seen like structural equation modeling where nodes are variables and edges are the correlation between or among variables conditioning on the structure of the graph. In the causal bayes graph the nodes are variables and the edges are statistical causal influences so causality and statistics is not always the same as what people mean by cause in the real world but like granger causality and just this notion of like coarse graining and cause it may line up with the sort of narrative natural language description that somebody has between when they're speaking it might be narrower than that it might be a different thing than that it might be more general than that. In a non-statistical interaction, I mean there's various ways to approach that it's kind of pointing towards some sort like a touch could be an interaction but what if things are touching but those variables don't change as a function of their touching then do those things interact or not that's one question and then cause there's like just tremendous real world and philosophy questions like what causes what what's the difference making cause what's the cause that makes a difference necessity insufficiency there's so many questions in this like causal philosophy and this is highlighting we're talking about the model we made a base graph with height weight and shirt color and here are the statistical effectors and so we're not saying that there's no interaction between this and that in the real world now we're within the model and we're talking about the causal graph and again unless the base graph is fully there's going to be some partitioning scheme that will result in some set of nodes being conditionally independent from another set of nodes when conditioned upon a blanket so it's not like features of the real world or even variables in a statistical equation can just be unilaterally tagged as blanket states it's always a partition that co-instantiates two sides internal and external but there's a symmetry and the blanket so it's not like features of the world or even features of the model are intrinsically internal external or but that's a partitioning scheme that can be applied to base graphs of like a vast various structures of bayes graphs and bayes graphs can be applied to a vast number of real world situations and many many papers and live streams discuss like sort of like all-encompassing markovian modeism markov blankets all the way down nested markov blankets are the world ranging to the critical perspectives not necessarily detractors but just those who believe that the usage of the concept is uh out of alignment with ways that people have used it. Mike and then anyone else so does it make sense to think of a markov blanket in the context of an internal agent state and an external environment is the markov blanket serving as this kind of a decoupling component because of the way that things are conditionally independent based on the variables inside the what do people think I mean for the internal state to be different than the

external state or have some other equilibrium other non-equilibrium state that the external environment is in you would have to necessarily have i'm not sure if decoupling is the right word but some sort of conditional dependent independent sort of zone that separates those two regions so here's figure 31 thank you brock the active and the sensory states compose the blanket now in various like live streams and papers the history of the concept again from the analytical pre-computational phrasing of insulation of fluctuation of random variables to pearls bayesian and computational framing however pearl's markov blanket concept doesn't have a delineation of active and sensory states it's only like there's one kind of fabric in that blanket it's just blanket states fristen maybe this is also will be shown to be traced to other places and ways but one of the core things that fristen at all have brought into the picture was interpreting within the blanket the states that have incoming dependencies to the system of interest as sensory and then the states that have outgoing dependencies from the system of interest as active states people may already see like challenges and complexities that arise with the arrows and the directions and a lot of that is addressed in the how particular is the free energy principle of aguilara at all and um we're going to continue talking about this entity partitioning ali and then lyle uh well i think uh we should point to a very important typographical error in picture here because uh in this figure active states are expressed in terms of external states and markup blankets and sensory states are expressed in terms of internal states and markup blankets but it should be vice versa i mean the flows of internal active states are independent of external states and the flows of external and sensory states do not depend on internal states so we should change x and μ in those two equations yes sometimes with a um possible typo like that it's like it's it's so egregious slash blatant that it's challenging to even know but we can absolutely um ask the authors but just to like read the equation and think about why it is that way the dot is the rate of change of a variable so the internal states are μ and so this is the rate of change of internal states rate of change of external states x and the rate of change of u the active states and y the sensory states so we're looking at how these partitioned states in a bayes graph change through time the states that are the blanket and the internal are the particular states those are referring to the particle that's like figure versus ground the particle that's moving around the curious particle the active entity is the particular states that is partitioning the thing literally the thing away from the niche then there's the blanket states that are intermediating that interface and then we'll also talk about autonomous states which are just the internal and the active states so the equation is saying the rate of change in for example active states is a flow function f and a noise function both the

flow and the noise function the subscript is referring to like that one it's kind of like ipso in latin like it's the flow on μ and the noise on μ the flow which is what the principle of least action converges us towards in the limit when the statistical fluctuations from the ω are low is a function of certain variables and so this is saying the flow of active states external sensory and active states however as ali has pointed to the arguments that should be guiding the flow of active states are actually the blanket states and internal states not external states and then analogously sensory states should have a flow that's defined in terms of blanket states and external states not internal states um lyell and then ali yeah so actually i think you're getting right right at the question at hand because i was confused about the the bi-directional arrows particularly between internal states and active states right and i think you're drilling right into that i still don't completely understand what that picture should look like uh but i i have trouble conceptually i sort of get a conception i feel like i get the markov blanket but that representation i have trouble working through the details on that cool so um guest stream number seven how particular is the physics of the free energy principle it explores different entity niche causal relationships like for example we could imagine a simple around-the-clock entity model external states induce sensory states one directional arrow sensory goes into internal internal to active active to external like an ooda loop or something like that but just in around the clock no bi-directionality no connection between active and sensory but given this partitioning one could imagine that topology of cause one could imagine all kinds of topologies of cause some that violate the markov blanket condition like internal states and external states that are um no longer internal and external because they would have a causal link but even preserving the markov blanket condition we can imagine different topologies different sparsities of couplings between these different states a bi-directional arrow is kind of like there would be a non-zero cause and in the other side of the matrix it would also be a non-zero cause versus a unidirectional relationship where like variable two has a causal influence on three but three doesn't have it on two in this paper um in guest room 7.1 and and that paper and further line of research they explore what sparse coupling structures causal architectures of the entity niche relationship and interface would grant different statistical capacities so um i could be wrong on this but active inference is not a commitment to this specific coupling architecture it may be possible to design or describe or imagine systems that have different causal architectures maybe even true at this basal most kernel loop but certainly true when we start thinking about nested structures and cognitive structures and so on um so yes it is a little unfortunate about this however um like this is just thinking about causal relationships of the

particular partition and we're describing statistical relationships here causal influence as inferred from like time series data with grainger causality or other methods on specific random variables so this is not the conversational notion of cause like but what i'm thinking is eventually resulting in things happening in the outside world or aren't things in the outside world eventually changing things yes but just like the chessboard through the blanket is statistically intermediating and again that's why it gets so messy with the application of like well the retina must be the sense if one can read that example not fall into the quicksand it could be didactic to tag retina as sense state is like the tip of the iceberg of a partially specified model often one where the measurements have not actually occurred and so on so then that can be extremely difficult for those with less familiarity with the formalism to generalize accurately from because shouldn't it be like you know retina sense states and then like the arm is the active state but it's a partitioning that co-instantiates and it's model dependence it's not describing features of the real world ollie uh yeah i just wanted to point uh one of ram said at all's um recent fascinating paper on bayesian mechanics which i put the link in the chat and in that paper especially in section three they have a very illuminating discussion about these whole business of uh partitioning and markup blankets and uh one sentence that's uh pretty relevant to our discussion here is that the key point to note is that the flow of internal and active components i.e their trajectory through state space does not depend upon external components and reciprocally the flow of external and sensory states or paths does not depend upon internal states or paths from page 15. so yeah in that if you refer to that article it has been elaborated much explicitly also in four days we will have dalton and maxwell at all for a guest stream the paper just came out now we'll be able to have a discussion with them and um learn and ask them questions so like if people have suggestions for guest streams that can be accomplished if people want to facilitate or participate in different streams if they have questions for different streams being there live but especially getting involved eagerly and actively and early is the biggest leverage point because that allows us to design the material so that it's like permanently useful rather than potentially ad hoc questions that arise during which can be super important and um hopefully people can see some affordances for participation and development um let's just get to this last question and it's it's like we have other questions but these are essential and it's one of the opening formalisms of the chapter just to conclude here though why does independence between the flows on the two sides of the blanket matter so much these are good things to think about um just one short thought would be without independence between the flows there is no distinguishing one thing from another

ali and also i think it relates to the uh what we uh read in chapter two about hidden states because uh that's what the word hidden probably refers to i mean internal state does not have uh direct access to external states and vice versa that markov blanket pretty much formalizes uh this hiddenness awesome and um partially observable models of the bayesian type which are used in like partially observable markov decision processes expectation maximization any kind of bayesian priors and hyper priors and so on all of these have bayes graphs that reflect conditionally independent so it's not that they're independent in a conversational way like they don't have any way of communicating it's that they are specifically involved in a causal network that has conditional independence but we'll hopefully develop more answers and notes here we'll move on what is the um differences between and among the terms bayesian brain hypothesis predictive processing predictive coding and active inference might you be able to suggest references or resources for where these distinctions are delineated sure so anyone can add more but here's i think two key resources here the first is livestream 43. in 43.0 maria gave a really excellent overview predictive processing and predictive coding from a historical perspective and briefly she proposed that predictive coding can refer to a uni-directional data encoding scheme of the kind that we see in signal processing video compression so on whereas predictive processing is referring to a bi-directional architecture where predictive coding is implemented in this ongoing top-down bottom-up way bayesian brain hypothesis was also touched on and we connected this to active inference but 43.0 is a long technical review paper but we have three discussions on it predictive coding theoretical experimental review recent paper very um good source for the um predictive processing and coding and predictive processing and coding initially was more of about a sensory framework however at the end of that paper they show okay now we're going to do predictive processing on action it's active so predictive processing about action is six of one and half dozen of another within a margin of reasonable approximation similar but there's also some key differences reliance on different formalisms but these are all things for us to explore and unpack as we work on the ontology and so on specifically predictive coding active inference in the bayesian brain this is an interview that a departed colleague and i did with carl fursten in 2018 and we specifically asked the bayesian brain hypothesis predictive coding and the free energy principle are often equated with one another you yourself has suggested that the three frameworks are variations of the same basic mechanisms so 2018 was a very different time active inference had not been delineated fep in the same way that it is coming to be now and there was a lot of other between that and now but he gives an extended answer so for people

who want to learn about that his extended answer and martin's restatements of the bayesian brain hypothesis are still some of the best places to find clarity on that issue does anyone have any other thoughts or questions on this just one other note while anyone can raise their hand is like the bayesian brain hypothesis sometimes um could be used in a a very um instrumentalist way we're using bayesian statistics to do neurobehavioral research it might be um a realist implementational um claim like the brain is doing bayesian statistics or something like bayesian statistics so this hypothesis floats kind of among the layers of mars analysis like implementational algorithmic computational and people do use it to mean different and in bayesian graphs in bayesian statistics it can be about perception or it could be about action so bayesian brain hypothesis depending on how somebody frames it and what specific models they're talking about might include for example all of these because these might be applied in a bayesian framework to talk about the brain ben yeah i mean i think you've covered it really really well but i i wish i'd have had access or knowledge of the result the the talk that you just mentioned because trying to differentiate predictive processing from active inference has been a kind of source of pain and confusion for me over the last few months and i just i thought another way of putting it perhaps because i think i know the difference now is essentially some presentations of predictive processing are active inference in the sense that they're highly embodied inactive embedded and so if you have a predictive processing that draws on these four reaccounts of cognition it tends to just that just is active inference active inference is kind of necessarily inactive and embodied and there's a paper um i should say there's a paper by i i believe it's called wilding the predictive brain it's by andy clark kate nave mark miller and george dean i think which basically takes predictive processing as a theory neural processing and weaves in these kind of cognitive these four e narratives really really well and kind of fleshes out a much more active infancy picture it's kind of the paper brings predictive process and active inference together really nicely i think thanks great suggestion and just without going into too much detail if this is the causal graph architecture of a predictive processing system one can imagine that like mu being a markov blanket with respect to these two epsilons or so on so predictive processing doesn't highlight as much the markov blanket formalism predictive processing is not a physics for particular systems like fep is it's not incompatible maybe they're just different fingers pointing at similar or different parts of the moon um so there isn't an incompatibility especially as over the recent years active inference implemented predictive architectures and approaches and predictive processing started to undergo that pragmatic inactive 4e5e whatever um infusion which led to them modeling action as a variable

and so all they did was they just made it so instead of the free energy being only about sensory observations and expectations they just tucked action in and it doesn't radically restructure the architecture it just means that action is a variable in these equations 43.1 uh zero one and two are all good places to look this was a short question internals estate and external state were added to the ontology but sometimes um like if you have a quotation mark and then an at it won't bring it up so sometimes you have to type it with an at first in our um last 15 minutes okay these two are um good maybe we could do a first pass on them and then in the coming week like add notes and more questions to discuss they wrote if one defines preferred states as expected states then one can say that living organisms must minimize the surprise of their sensory observations i'm not clear on the ontology here is it that preference is the same as expectation or that formally we can use them in the same equation or something else does active inference make ontological claims like preference is actually expectation just as heat is actually the excitation of molecules more generally is the free energy in physics just an analogy or is it an ontological assertion who would like to approach one of the aspects of this question i can i i would like to take a stop but again i guess guess yes please i guess it's not saying that they're the same but that we want like if we're minimizing free energy we're trying to make it the same and but if they're surprised they won't be the same i guess the goal is to get them as close as possible that's that will be my guess on this good insight um also point to two ways that expectation is conversationally used um the expectation with the fancy e is the expected mean of a distribution like expected returns might be about the future but it's actually referring to the mean estimate of a distribution in the gaussian case the mean and the mode are the same mode seeking and mean seeking are equivalent there can also be a variance estimator and so on so the expectation of a gaussian is like tremendously informative expectation can also mean things that are predicted about the future that's a conversational what do you expect it to be tomorrow both can be conjoined like what do you expect the temperature to be tomorrow is the expectation of a distribution of the uncertainty of the temperature distribution at the time point tomorrow so they're not exclusive definitions it just has to be understood what this means because it'd be like well if our predictions about the future are our preferences you know then there's all kinds of tangles that you know one might find themselves in the um i'm uh not remembering which number it was but um the dual usage of the same term to reflect the organism's preferred and also the centering of the distribution at that time point and other time points and then just like jessica mentioned operation of free energy minimization is in service of reducing the divergence between the preferred state and the um the expectation of

the preferred state which is can be set or learned and then the um the observations that are coming in in one equation it leans more towards the english description of a preference like the organism prefers to be at 72 and in the other sense that is like the expectation of a distribution and there's some observation distribution with a minimization of a divergence such that the observations were realizing preferences then those distributions would be aligned so yes these terms are not these terms are more like phenomena or natural language tags also it'll be awesome in the last two weeks of june in 46 active inference models do not contradict folk psychology this will be extended extended discussions on like wanting beliefs intentions which variables inactive are appropriate for being described that way does it contradict one or the other those are like things that we'll explore the equations are just what they are and then in some usages a variable is like the organism's preferred states what it's trying to reduce its divergence and also that is like where it expects to be and that's what licenses the use of surprise and minimizing surprise because an observation right at the center of the gaussian is minimally surprising we talked about that in chapter two an observation that's outside of the gaussian is highly surprising if we have some sort of distribution outcomes we could ask how closely are they aligned with our preferences in terms of surprise due to this dual use of the same claims do theories make claims do people make claims eg preference is actually expectation just as heat is actually excitation of anyone can raise their hand at any time we'll talk about that in the folk psychology but it's an interesting question ben yeah i wonder if um it seems to me like these questions are taking us maybe into the map territory um debate that we i think we spoke about last time about the you know how um realist or concrete should we think the kind of ontological claims of active inference there's certainly certainly the last part of that question i think about whether we should take it as an analogy so yeah i think it's quite a big question but i don't i don't know okay i won't find the slide here but um is the free energy in physics an analogy not sure if this is saying like the gibbs free energy or if that's actually talking about like the variational free energy or the expected free energy which are statistical so i don't know if it's an analogy or an ontological assertion other than just to connect it to like a linear model with a gaussian error distribution is that an ontological assertion that there is a gaussian error distribution in the world no are statistical quantities ontological assertions about the world maybe there's some nuance but broadly speaking no and also just like on the math learning group we've expanded our operation due to many active participants so just note that the meetings are at 19 utc on monday tuesday wednesday and thursday and they're in the discord voice chat not in this gather we've been working on the resources on the notation basic

and background questions as well as overviews of the math for different chapters like basically just summarizing several um people and this is an example of like the broken link it's like variational free energy is deleted so just one time somebody has to come through and just add it back in but these are natural language descriptions that will be very easy to um transmute into computer code and translate amongst human languages and provides a lot of legibility and comprehensibility so a lot of the math group has been modifying like the equations and everybody is welcome to join those like discord sessions and contribute with questions or with knowledge and expertise like no matter where somebody is in learning or discussing these equations they are the essence of active inference so questions random notes related thoughts expertise are all essential for us improving our shared understanding and then just in the last five minutes um page 42 they wrote in advanced organisms preferred states can also extend to learned cognitive goals and go on to say that advanced organisms like human can achieve preferred states by increasingly abstract social cultural strategies like they talked about thermoregulation and then all the way up to like air conditioning distribution systems my question is if they are cast in terms of active must all of these preferred states and strategies be ultimately linked to survival in some way or in the case of advanced organisms can free energy be related to something other than survival yeah it feels like this notion of survival and preferred states gets conflated in a sort of odd way not everything is about survival right and so you might have some measurement of surprise if you're in a temperature that's 120 degrees but it's survivable for a period of time whereas if you're in a temperature of 300 degrees fahrenheit not survivable right but if you contrast that with other things that could be equally surprising or outside the distribution as you described it earlier the notion of that all swans are white but then you find a black swan and so uh you'd never conceived that there might be a black swan before that lives outside your distribution and creates a surprise but it's not necessarily survival rated thank you so there's a there's a sort of a missing severity in there somewhere okay thank you ali and then lyle well i think work life here is used somehow in a more black way meaning system that resists dispersion and energy dissipation so it's not necessarily biological survival yes great point systems that fail to persist survive will not continue to be that thing and that's just in like a physical sense repeated measurements are only enabled by systems that are persistent at that time scale not eternally and so when we want to have a theory or a framework or a physics for things then they have to look as if they are minimizing free energy otherwise they will simply not be that thing ben and then lyle yeah so this was my question i suppose just to just to kind of clarify it quickly um it just one of the

things i wonder most about active influences it seems to me at least in my life i'm quite often making decisions and having preferences that seem so abstracted from you know my attunement with my environment or my continued viability as a system you know like my preference which mug i'm gonna drink coffee out of in the morning it seems in some sense removed my free energy the way that free energy is described in the literature um so that's kind of what i meant great in um uh yesterday's 45.2 we asked specifically like how can all behavior be optimal when like mistakes are made or oddly ask the question like what about when there's like sacrificial or alt altruistic behavior and there's a great answer in 45.2 but suffice to say that with different priors all behavior can be cast as optimal behavior so the normativity isn't referring to what should be done or what is best but it's rather like a process normativity lyle right so yeah this for me tees up the question i continue to have about the approach which is how does how does the concept of novelty which organisms would pursue fit into this concept of i mean how would it even be represented here and and novelty is not just a human trait but you would see it across all kinds of creatures that would seek out would seek out novel and surprising experience you know not expected so i for me it's it's kind of an open question about how that's captured in this framework yes the beginning and ending of 45.2 carl's um focus was on curiosity there's various ways to approach it and it's an actual question about novelty of different types and scales within the hierarchical modeling one can understand novelty of different kinds and this again is just like the kernel this is like the single linear regression and then there's like model selection across hierarchically nested linear models much development beyond the kernel but this is like a kernel but um and especially when thinking about expected free energy a path can move to a novel area because it's part of an expected trajectory that does reduce risk and ambiguity and so on so that's the deflationary approach to novelty pursuit jessica and then we'll end the discussion at the end danny you mention it and i guess i i see novelty i sort of like the epistemic origin and so it's like you are seeking on like new information new in order to minimize uh that uncertainty i not so much that novelty is on cellar than t because like like if we look at no i've been thinking like oh it's new and therefore it's going to be surprising and and so we want to minimize those things and then yes or also if we look at uncertainty equal to risk we can say like novelty is because of the unknown represented risk and if we're minimizing uncertainty we're trying to minimize rest but from what i saw a little bit on chapter two and it's sort of like we're not necessarily trying to minimize race or like the unknown and like depending on our preferences we might tolerate high res and what we're trying to minimize is the surprise of what we expect like if i'm in

finance and i expect very high risk in an investment but i already know that you know and high risk comes back on like my um uncertainty was minimized even though there was a lot of risk or there were a lot of like unknowns on or like novelty and things like that but it was already in my expectation of what was going to happen i was already minimizing that even in reality it may look like um that there was a lot of uncertainty uh in actuality um but in my model there isn't and so that's kind of like how i explained that and to me a little bit and in terms of novelty for like adventure and like on curiosity and things like that i i equated more to epistemic foraging as a way of learning and updating the model and that's why we're doing it thanks jessica i'll just add one closing note on that i would argue that it's the pragmatic absolutism the reinforcement and reward learning centrism that has curtailed an effective theory of novelty or curiosity radically different kinds of actions have to be coerced into a value framework whether it's deep q-learning or reinforcement or anything like that there has to be like a novelty bonus or all these other ad-hoc unprincipled approaches they might be super technical they might be super effective but they're not driven from first principles in variational free energy f and expected free energy g there's this like minimum of two with pragmatic value which is to um bring alignment between the preferences and the observations and the epistemic value associated with what is casually called novelty or curiosity at that scale of potentially a hierarchical model so active inference does provide partitioning and an imperative that helps us address the phenomena of curiosity and of novelty in a way that is quite disparate from trying to coerce it into a value of curiosity is expected pragmatic value or the value of basic research is the expected utility that it could bring so but these are all really awesome and open and ongoing questions so we will um close the recording and then have just a one minute break then in this room we'll continue with dot tools organizational unit and if anybody wants to like talk about something else they can go to a different um gather room so thanks everybody and see you next week thanks